

Improving FAA Academy Training Outcomes for ATC
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Executive Summary

The FAA Academy is central to addressing air traffic controller staffing challenges faced by the agency. While past improvement efforts have focused on increasing throughput, modernizing technology, and improving efficiency, true efficiency in a training system cannot be achieved through acceleration alone. Achieving efficiency, and ultimately successfully proficient performance, depends on how well the Academy is aligned with downstream training and real-world operational demands. Discussions with both field controllers and Academy instructors reveal a lack of shared expectations around the proficiency level of Academy graduates. The Academy is not intended to produce fully certified controllers, but clearer alignment between Academy standards and field expectations is essential and should be consistently defined and communicated broadly. The Academy cannot, of course, produce near-certified controllers, but it can develop a level of readiness that maximizes each trainee's likelihood of success. Efforts to improve Academy training should focus on aligning the readiness it delivers with the preparation required to succeed in field training.

Achieving this level of alignment is challenging, but the issues can be organized into three high-leverage areas that work together to improve consistency, clarity, and continuous learning across the training system. First, performance measurement and expectations must be strengthened so trainee performance is recorded, observed, and interpreted in a consistent and reliable way across training stages. Second, readiness signaling at transition points must be clearly defined and standardized, so that readiness is consistently understood, measured, and communicated across training environments. Third, downstream performance must be used to inform training earlier by systematically capturing how trainees perform in the field and using that information to refine training at the Academy. Together, these areas create a more connected system that improves decision-making, reduces variability, and increases the likelihood of successful training outcomes.

This paper outlines three high-leverage areas to address these challenges:

1. **Performance Measurement and Expectations.** Strengthen and standardize how trainee performance is recorded, observed, measured, and interpreted, particularly in complex scenarios involving workload, prioritization, timing, and decision-making under pressure. While some assessments can be easily standardized (e.g., knowledge test), more complex evaluations that rely on human judgement often introduce variability. This inconsistency can reduce reliability and make it harder to compare performance across trainees and training stages. Improving the structure and consistency of performance observation, with clearer links between tasks, expectations, and outcomes, supports more reliable identification of emerging difficulties, enables earlier and more targeted remediation, and increases confidence in progression decisions.
2. **Clarify and Standardize Readiness Signaling at Transition Points.** Clarify what readiness represents and ensure it is consistently defined, measured, and communicated across and between training environments. A trainee may meet Academy standards, but variability in how that readiness is interpreted can create uncertainty for facility instructors and require additional time to reassess capability. Strengthening readiness signaling means moving beyond a general "ready to progress" determination to a clearer, evidence-based representation of performance. For example, instead of simply indicating that a trainee passed, readiness could include structured information such as performance in high-workload scenarios, ability to prioritize and recover from errors, consistency across simulations, and areas where additional support may be needed. This type of information provides a more complete picture of how the

trainee is expected to perform under operational conditions. When readiness is defined using consistent, observable indicators and communicated in a structured way, it reduces ambiguity, supports more efficient and predictable downstream training, and improves alignment between Academy outputs and field expectations.

3. **Use Downstream Performance to Improve Upstream Training.** The clearest test of Academy preparation occurs in the field, where trainees must apply their skills under real operational conditions. While trainees may meet Academy standards, some still struggle with workload, prioritization, timing, or decision-making when complexity increases, revealing gaps between training outputs and field expectations. These gaps are not always consistently captured or used to inform earlier training, which allows similar issues to persist across trainee cohorts. Strengthening this feedback loop means systematically capturing where trainees struggle in the field and linking those challenges back to specific training elements. For example, if trainees consistently have difficulty managing workload during high-traffic scenarios, that pattern can be used to adjust simulation design, performance expectations, or instructional emphasis at the Academy. Using downstream performance this way enables targeted improvements, reduces recurring gaps, and ensures training is aligned with operational demands. Without this connection, issues are addressed later in training rather than prevented earlier.

Building on these three high-leverage areas, the following recommendations focus on strengthening how performance is measured, how readiness is defined and communicated at transition, and how downstream performance is used to continuously improve training. Together, they are designed to create a more connected and data-informed system by targeting the underlying mechanisms that drive consistency, clarity, and learning.

Key Recommendations

- Establish a system-level approach to integrate trainee performance data across training stages.
- Define readiness with standardized, observable performance indicators that clearly reflect expectations across training environments. This includes metrics, structured tracking, shared visibility, alignment to objectives.
- Standardize and operationalize readiness using observable performance-based indicators. This includes definition, communication, readiness profiles.
- Align readiness and simulation with operational demands. This includes high-variability scenarios, and decision-making under uncertainty.
- Establish structured feedback mechanisms between the academy and field facilities. This includes downstream data use, gap identification, training improvement.
- Use downstream performance information to inform training improvements. This includes error tracking, early indicators, and reducing unnecessary delays vs. Necessary time.

Key Takeaway

Improving efficiency at the Academy is less about increasing training speed and more about ensuring strong alignment with downstream demands, clear and consistent performance measurements, and effective use of performance data to drive continuous system improvement.

When performance is clearly understood, readiness is consistently defined, and feedback is shared across training stages, the system becomes more efficient, outcomes become more predictable, and the likelihood of certification success improves. Strengthening the Academy's role as a central control point within this system is one of the most effective ways to improve both safety and workforce outcomes.

Appendix A

Performance Measurement and Expectations				
Recommendation	Rationale	Alignment with GAO Report	Alignment with FAA 2026 Flight Plan	Impact
Establish a system-level approach for integrating and using trainee performance information across training stages.	Enables full system visibility and consistent decision-making	End-to-end pipeline visibility	People: Identify and reduce training bottlenecks; Safety: Support integration with the Safety Management System (SMS)	Reduces fragmentation, improves coordination, increases throughput
Implement structured, objective performance tracking supported by standardized metrics to improve evaluation consistency, reduce variation in instructor interpretation through evaluator training and consensus building sessions, as well as using multiple raters.	Improves evaluation consistency by reducing interpretation variability and enabling more reliable decision-making at all levels.	Standardized evaluation and performance accountability	Safety: Support risk-informed decision-making and transparency; People: Provide improved training tools	Enables more consistent progression decisions and improves leadership visibility for informed resource allocation.
Expand simulation training to better reflect operational demands.	Improves transfer of training to real conditions	Training effectiveness	People: improve training quality; NAS Modernization: realistic environments	Faster adaptation, improved readiness
Prioritize scenarios that reflect high-variability and transition-critical conditions requiring real-time decision-making under uncertainty.	Focuses training on high-risk conditions where failures occur and builds the cognitive skills needed to perform under pressure and complexity.	Targeted training improvements and operational readiness	Safety: Support proactive risk identification and hazard mitigation	Reduces breakdowns in critical scenarios and improves performance under pressure.

Align lessons, simulations, and evaluations with clear learning objectives to improve consistency, comparability, and diagnostic accuracy.	Ensures alignment between training content and performance expectations, enabling more precise measurement and diagnosis.	Measurable outcomes	People: improved training structure	Clearer performance tracking, targeted remediation
Implement structured error tracking and performance trend analysis to identify recurring patterns, detect early indicators of instability, and understand where performance breaks down as complexity increases.	Identifies root causes of performance breakdowns and enables early detection of instability through performance trends as task complexity increases.	Root cause analysis and proactive risk management	Safety: Support risk identification and mitigation through proactive hazard management	Leads to more effective fixes, reduces recurring errors, and lowers attrition and late-stage failure.
Clarify and Standardize Readiness Signaling at the Point of Transition				
Recommendation	Rationale	Alignment with GAO Report	Alignment with FAA 2026 Flight Plan	Impact
Standardize and operationalize readiness using observable, performance-based indicators that reflect expectations across training environments and ensure consistent communication of performance information from the Academy to facilities.	Reduces ambiguity at transition by making readiness clearly defined, measurable, and consistently applied across training environments.	Standardization across the training pipeline and establishment of measurable performance expectations	People: Reduce inefficiencies in the training pipeline; Safety: Ensure consistent, standards-based evaluation	Enables faster transitions, reduces rework, and improves clarity, consistency, and comparability of performance assessments.
Train instructors and evaluators to apply performance standards consistently, calibrate through practice and consensus, and monitor rating reliability within the training data system.	Improves evaluation consistency and accuracy by reducing interpretation variability and enabling shared understanding of performance standards.	Standardized evaluation and performance accountability	Safety: Support risk-informed decision-making and transparency; People: Improve training tools and strengthen evaluator capability	Leads to more reliable and defensible performance assessments, reduces variability across instructors, and increases confidence in progression and certification decisions.

Align readiness with real operational demands, so it reflects the ability to apply skills under workload and complexity similar to field training.	Ensures readiness reflects real-world performance demands and operational conditions.	Training relevance	Safety: Align with real-world risk conditions	Improves downstream performance and stability
Use structured, objective performance evidence to develop consistent and shareable readiness profiles.	Improves continuity across stages	Information sharing	People: improved coordination across system	Reduces diagnostic burden, increases efficiency
Reduce early diagnostic burden in facility training by providing clearer trainee performance information.	Eliminates redundant assessment	Pipeline efficiency	People: resolve training bottlenecks	Speeds up training progression
Track transition performance as a core system indicator.	Identifies alignment gaps at the transition point and enables early detection of mismatches between Academy preparation and field expectations.	Cross-stage outcome tracking	Safety: performance monitoring	Leads to earlier identification of misalignment, reduces rework, and improves transition efficiency and downstream performance stability.
Use Downstream Performance to Improve Upstream Training				
Recommendation	Rationale	Alignment with GAO Report	Alignment with FAA 2026 Flight Plan	Impact
Track trainee performance across training stages to link Academy and field outcomes and understand how early capability affects adaptation, stabilization, and certification.	Links performance across training stages to show how early capability influences downstream outcomes.	Lifecycle analysis	People: workforce pipeline optimization	Leads to improved understanding of training effectiveness and supports more targeted, system-level improvements.
Establish structured feedback mechanisms between the Academy and field facilities.	Enables continuous learning by systematically capturing and sharing performance information across training environments.	Coordination across system	People: improve training processes; Safety: learning system	Leads to improved training design, reduces learning gaps, and strengthens system-wide coordination.

Identify recurring gaps between Academy training and operational performance to guide targeted improvements in training and evaluation.	Identifies recurring gaps between training and operational performance and enables targeted improvements based on observed field challenges.	Systemic issue identification	Safety: hazard identification	Leads to reduced repeated deficiencies and improves alignment between training and operational demands.
Enable shared visibility of trainee performance across training stages to improve alignment and reduce transition uncertainty.	Enables shared access to performance information to support alignment, coordination, and system-wide understanding.	Enterprise data integration	People + Safety + NAS Modernization: data-driven system	Leads to improved alignment across training environments, reduces transition uncertainty, and strengthens system-level decision-making.
Use downstream performance information to inform training improvements.	Ensures training improvements are grounded in real-world performance outcomes.	Evidence-based improvement	Safety: continuous improvement; People: training refinement	Improves training relevance and effectiveness